Computer Organization & Assembly Language

Revision: Data Representation

Number System

- Any number system using a range of digits that represents A specific number. The most common numbering systems are decimal, binary, octal, and hexadecimal.
- Numbers are important to computers
  - Represent information precisely
  - Can be processed
- For example:
  - To represent yes or no: use 0 for no and 1 for yes

Decimal Number System

- A numbering system that uses ten digits, from 0 to 9, to represent numerical values/quantities. Each digits has a weighted value of $10^0, 10^1, 10^2, 10^3$ and so on, ranging from right to left.

Binary Number System

- A numbering system that uses two digits 0 and 1, to represent numerical values/quantities. Each digits has a weighted value of $2^0, 2^1, 2, 2^3$ and so on, ranging from right to left.

Hexadecimal Number System

- A numbering system that uses sixteen digits, from 0 to 9 and A to F, to represent numerical values/quantities. Each digits has a weighted value of $16^0, 16^1, 16^2, 16^3$ and so on, ranging from right to left.

Conversion Between Number Systems

- Converting Hexadecimal to Decimal
- Multiply each digit of the hexadecimal number from right to left with its corresponding power of 16.
- Convert the Hexadecimal number **82ADh** to decimal number.
- 0x82AD (memory addressing)

Conversion Between Number Systems

- Converting Binary to Decimal
- Multiply each digit of the binary number from right to left with its corresponding power of 2.
- Convert the Binary number **11101** to decimal number.

Conversion Between Number Systems

- <u>Converting Decimal to Binary</u>
- Divide the decimal number by 2.
- Take the remainder and record it on the side.
- REPEAT UNTIL the decimal number cannot be divided into anymore.

Conversion Between Number Systems

- <u>Converting Decimal to Hexadecimal</u>
- Divide the decimal number by 16.
- Take the remainder and record it on the side.
- REPEAT UNTIL the decimal number cannot be divided into anymore.

Binary Arithmetic Operations

- Addition
- Like decimal numbers, two numbers can be added by adding each pair of digits together with carry propagation.

11001
+ 10011
101100

Binary Addition

647
+ 537
1184

Decimal Addition

Binary Arithmetic Operations

- Subtraction
- Two numbers can be subtracted by subtracting each pair of digits together with borrowing, where needed.

11001
- 10011
00110

Binary Subtraction

627
- 537
090

Decimal Subtraction

Hexadecimal Arithmetic Operations

- Hex Addition
- Like decimal numbers, two numbers can be added by adding each pair of digits together with carry propagation.

5B39
+ 7AF4
D62D

Hexadecimal Addition

HexaDecimal Arithmetic Operations

- Hex Subtraction
- Two numbers can be subtracted by subtracting each pair of digits together with borrowing, where needed.

D26F
- BA94
17DB

Hexadecimal Subtraction

MSB and LSB

- In computing, the most significant bit (MSB) is the bit position in a binary number having the greatest value. The MSB is sometimes referred to as the left-most bit.
- In computing, the least significant bit (LSB) is the bit position in a binary integer giving the units value, that is, determining whether the number is even or odd. The LSB is sometimes referred to as the right-most bit.

Unsigned Integers

- An unsigned integer is an integer that represents a magnitude, so it is never negative.
- Unsigned integers are appropriate for representing quantities that can never be negative.

Signed Integers

- A signed integer can be positive or negative.
- The most significant bit (MSB) is reserved for the sign:
    - 1 means negative and 0 means positive.

- Example:

    00001010 = decimal 10

    10001010 = decimal -10

One’s Complement

- The one’s complement of an integer is obtained by complementing each bit, that is, replace each 0 by a 1 and each 1 by a 0.

2’s Complement

- Negative integers are stored in computer using 2’s complement.
- To get a two’s complement by first finding the one’s complement, and then by adding 1 to it.

- Example

    11110011 (one's complement of 12)

    \+ 1 (decimal 1)

    11110100 (two's complement of 12)

Subtract as 2's Complement Addition

- Find the difference of 12 – 5 using complementation and addition.
- 00000101 (decimal 5)
- 11111011   (2's Complement of 5)

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          00001100   (decimal 12)
        + 11111011   (decimal -5)
No Carry  00000111   (decimal 7)
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Example

- Find the difference of 5ABCh – 21FCh using complementation and addition.
- 5ABCh = 0101 1010 1011 1100
- 21FCh = 0010 0001 1111 1100
- 1101 1110 0000 0100 (2’s Complement of 21FCh)

  0101 1010 1011 1100 (Binary 5ABCh)

+ 1101 1110 0000 0100 (1’s Complement of 21FCh)

10011 1000 1100 0000

Discard Carry

Binary, Decimal, and Hexadecimal Equivalents.

Binary	Decimal	Hexadecimal		Binary	Decimal	Hexadecimal
0000	0	0		1000	8	8
0001	1	1		1001	9	9
0010	2	2		1010	10	A
0011	3	3		1011	11	B
0100	4	4		1100	12	C
0101	5	5		1101	13	D
0110	6	6		1110	14	E
0111	7	7		1111	15	F